

L5b: Multiple Asset Geometric Brownian Motion

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Risk. Trading involves risk and is generally not recommended for those with limited resources, experience, or a low risk tolerance. Past performance does not guarantee future success.

Responsibility. You are responsible for your investment decisions based on your financial circumstances, objectives, risk tolerance, and liquidity needs.

Objectives for Today

By the end of this lecture, you will be able to:

- Update and extend GBM: update single asset GBM parameters with an exponential moving average, then model several assets whose prices move together.
- Estimate and interpret covariance: calculate the covariance matrix from historical growth rates, interpret covariance and correlation, and convert it to the model's covariance rate.
- Compare portfolio allocations: relate weights to buy-and-hold wealth, generate long-only allocations with the Dirichlet distribution, and compare their estimated growth and risk.

Lectures and Examples

Lecture: Multiple Asset Geometric Brownian Motion

Example: Updating GBM Parameters with EMA
Carried over from L5a. Can recent observations improve the forecasts? Update mean growth and volatility, then compare the forecasts with the frozen model.

Example: Covariance and Correlation from Data
How do firms' growth rates move together? We estimate covariance and correlation from historical data and connect them to the price model.

Example: Portfolio Weights and the Dirichlet Distribution
How does the investment mix affect growth and risk? We explore portfolio weights with the Dirichlet distribution and follow buy-and-hold wealth.

Concept Review: Updating GBM Parameters

An exponential moving average (EMA) updates the GBM estimates as new prices arrive, giving recent growth rates $g_k = \ln(S_k/S_{k-1})/\Delta t$ more weight.

Initialize at entry index s from the training estimates:

$$m_s = \widehat{\mu}_{g,0}, \qquad v_s = \frac{\widehat{\sigma}_0^2}{\Delta t}.$$

With decay factor $0 < \lambda < 1$, update at each new observation $k > s$:

$$\begin{aligned} \delta_k &= g_k - m_{k-1}, && \text{(innovation)} \\ m_k &= m_{k-1} + (1-\lambda)\delta_k, && \text{(mean update)} \\ v_k &= \lambda\big[v_{k-1} + (1-\lambda)\delta_k^2\big]. && \text{(variance update)} \end{aligned}$$

EMA: Recover the GBM Parameters

For the growth rate, the estimated mean m_k has units of year^{-1} and the estimated variance v_k has units of year^{-2}. Convert these moments to GBM parameter estimates:

$$\hat{\mu}_{g,k} = m_k, \qquad \text{mean growth rate,}$$

$$\hat{\sigma}_k = \sqrt{v_k \Delta t}, \qquad \text{volatility,}$$

$$\hat{\mu}_k = \hat{\mu}_{g,k} + \frac{\hat{\sigma}_k^2}{2}, \quad \text{drift.}$$

We estimate the volatility parameter by multiplying the growth-rate standard deviation $\sqrt{v_k}$ by $\sqrt{\Delta t}$. The result has units of year$^{-1/2}$.

Variance update: the recursion accounts for the changing mean; it does not include a finite-sample unbiased correction.

EMA: Interpret the Half-Life

The half-life $H_{\text{half}} > 0$ is the number of observations needed to halve the weight of an older observation:

$$\lambda = 2^{-1/H_{\text{half}}}.$$

With a half-life of 21 observations, $\lambda \approx 0.9675$. The updated mean combines 96.75% of the previous mean with 3.25% of the new growth rate.

A shorter half-life makes the estimates respond faster but also makes them more sensitive to noise. The half-life controls the parameter updates; the holding period sets the forecast horizon. Each forecast holds its parameter estimates fixed.

Derivation: Derive the exponentially weighted moments and GBM parameter updates

Example: Updating the Parameter Estimates

Example: Compare frozen and updated estimates

Can giving recent observations more weight improve our forecasts?

Each day, we update the probability that the original trade will exceed its NPV target at the end of the chosen forward window. We compare frozen parameters with EMA estimates of volatility alone or of both mean growth and volatility.

We evaluate the three forecasts using the coverage and width of their 95% growth-rate prediction bands.

Next, we turn to several firms whose prices move together.

Company Profile: Bridgewater Associates

Bridgewater Associates is a global macro investment firm that manages portfolios for pension funds, endowments, and sovereign wealth funds. Ray Dalio founded it in 1975. The firm tests cause-and-effect rules for economies and markets against historical data and builds portfolios around how assets move together.

- All Weather (1996): stocks, nominal bonds, inflation-linked bonds, and commodities respond differently to the same news, so the portfolio balances risk across them rather than dollars (risk parity).
- Pure Alpha (1991): the actively traded portfolio aims for returns uncorrelated with the stock and bond markets. *Alpha* is the return earned from skill rather than from holding the markets.
- Culture: radical truth and radical transparency, described in Dalio's *Principles: Life and Work*.

Bridgewater Associates: Explore Further

Jobs and internships. Browse the job openings and students pages. The 2027 Investment Associate Internship Program, an eight-week summer program in Westport, Connecticut, is posted as of September 21, 2026.

Watch:

- How The Economic Machine Works, a thirty-minute animated explanation of credit and debt cycles narrated by Dalio (Principles by Ray Dalio channel).
- The firm's own Bridgewater Associates channel.

Connection to today: a portfolio's risk depends on how its assets move together. Today we estimate that co-movement from data, build it into the price model, and see how the allocation changes portfolio growth and risk.

Multiple Asset Geometric Brownian Motion

Independent GBM models treat assets' growth rates as unrelated. Multiple asset GBM (MAGBM) lets them move together.

For assets $i \in \mathcal{P} = \{1, \dots, M\}$ with prices $S_i(t) > 0$:

$$\frac{dS_i(t)}{S_i(t)} = \underbrace{\mu_i \, dt}_{\text{drift}} + \underbrace{\sum_{\ell=1}^{M} A_{i\ell} \, dW_\ell(t)}_{\text{correlated noise}}.$$

- Shared inputs: $dW_1, \dots, dW_M$ are independent Wiener increments, each with mean zero and variance dt.
- Loading matrix $\mathbf{A}$ (year$^{-1/2}$): row i sets how strongly asset i responds to each input.
- Covariance rate $\mathbf{C} = \mathbf{A}\mathbf{A}^\top$ (year^{-1}): its diagonal holds the squared volatilities $C_{ii} = \sigma_i^2$.

Why $\mathbf{A}\mathbf{A}^\top = \mathbf{C}$?

For two assets, both noise terms use the same two inputs:

$$\text{Asset 1: } A_{11}\,dW_1 + A_{12}\,dW_2, \qquad \text{Asset 2: } A_{21}\,dW_1 + A_{22}\,dW_2.$$

Independent increments have $\mathrm{Cov}(dW_\ell, dW_k) = dt$ when $\ell = k$ and zero otherwise, so only matching terms contribute:

$$\mathrm{Cov}\Big(\sum_\ell A_{i\ell}\,dW_\ell,\ \sum_k A_{jk}\,dW_k\Big) = \sum_{\ell=1}^{M} A_{i\ell}A_{j\ell}\,dt = (\mathbf{A}\mathbf{A}^\top)_{ij}\,dt = C_{ij}\,dt.$$

Thus, C_{ij} is the covariance per unit time of assets i and j.

Choosing a Covariance Factor

Many loading matrices satisfy $\mathbf{A}\mathbf{A}^\top = \mathbf{C}$. Two standard choices are:

- Cholesky factor: when $\mathbf{C}$ is positive definite, choose the lower triangular factor $\mathbf{L}$ with positive diagonal entries:

$$\mathbf{C} = \mathbf{L}\mathbf{L}^\top, \qquad \mathbf{A} = \mathbf{L}.$$

- Symmetric square root: for a positive semidefinite $\mathbf{C}$, including a singular covariance matrix, choose:

$$\mathbf{A} = \mathbf{C}^{1/2}, \qquad \mathbf{A}\mathbf{A}^\top = \mathbf{C}^{1/2}\mathbf{C}^{1/2} = \mathbf{C}.$$

Both choices give covariance $\mathbf{C}\,dt$ for the model's random increments.

The Exact One-Step Transition

As in L4b, Itô's lemma gives each asset its own mean growth rate:

$$\mu_{g,i} = \mu_i - \frac{C_{ii}}{2}.$$

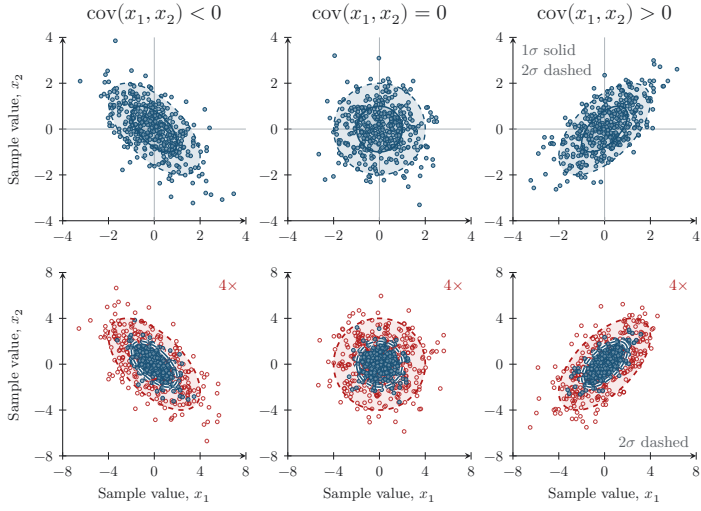
Draw M independent standard normal values $\mathbf{Z}$. The exact update over $\Delta t > 0$ years is:

$$S_i(t+\Delta t) = S_i(t)\exp\Big[\underbrace{\mu_{g,i}\,\Delta t}_{\text{mean log growth}} + \underbrace{\sqrt{\Delta t}\,(\mathbf{AZ})_i}_{\text{correlated fluctuation}}\Big].$$

- Within a step: use the same $\mathbf{Z}$ for all assets to preserve their covariance.
- Across steps: draw a new independent $\mathbf{Z}$ at every step.

Derivation: Itô's lemma for several noise sources and the scaling rule

Covariance: Direction and Scale



Empirical Covariance Matrix

Use $N \geq 2$ equally spaced periods of growth-rate data for M firms. Align the records so every firm's period k covers the same interval.

Let $g_k^{(i)}$ be firm i's growth rate in period k. Its sample mean is:

$$g_i' = \frac{1}{N} \sum_{k=1}^{N} g_k^{(i)}.$$

The empirical growth-rate covariance matrix $\hat{\Sigma}_g \in \mathbb{R}^{M \times M}$ has entries:

$$\hat{\Sigma}_{g,ij} = \frac{1}{N-1} \sum_{k=1}^{N} \underbrace{(g_k^{(i)} - g_i')}_{\text{deviation for } i} \underbrace{(g_k^{(j)} - g_j')}_{\text{deviation for } j}, \qquad i, j \in \mathcal{P}.$$

Each entry pairs observations from the same period and has units of year^{-2}.

Interpreting the Covariance Matrix

The covariance matrix describes individual growth-rate variability and how growth rates move together across firms:

- Diagonal entries: $\hat{\Sigma}_{g,ii}$ is firm i's growth-rate variance. Its square root is the growth-rate standard deviation σ_g for that firm (year^{-1}).
- Off-diagonal entries: positive values mean the two growth rates tend to deviate from their means in the same direction; negative values mean opposite directions.

Covariance also depends on the scale of each growth-rate series. Next, correlation lets us compare relationships on a common scale.

Correlation

For firms i and j with positive sample variances, correlation divides growth-rate covariance by the product of their standard deviations:

$$\rho_{ij} = \frac{\hat{\Sigma}_{g,ij}}{\sqrt{\hat{\Sigma}_{g,ii}\,\hat{\Sigma}_{g,jj}}} \in [-1, 1] \quad \Longleftrightarrow \quad \hat{\Sigma}_{g,ij} = \rho_{ij}\sqrt{\hat{\Sigma}_{g,ii}\,\hat{\Sigma}_{g,jj}}.$$

This dimensionless ratio lets us compare pairs of firms:

- Direction and strength: the sign gives the direction; values closer to -1 or 1 indicate a stronger linear relationship.
- Zero correlation: no linear relationship in this sample; this does not mean the growth rates are independent.

The Data Matrix and the Sample Covariance

Arrange aligned growth rates in $\mathbf{G} \in \mathbb{R}^{N \times M}$: rows are periods and columns are firms. Compute the covariance matrix in two steps:

- Center each column: let $\mathbf{g}' = (g'_1, \ldots, g'_M)^\top$ be the sample-mean vector and $\mathbf{1} \in \mathbb{R}^N$ a vector of ones. The outer product repeats these means in every row:

$$\tilde{\mathbf{G}} = \mathbf{G} - \mathbf{1}\,\mathbf{g}'^\top.$$

- Multiply the centered columns: the empirical covariance is:

$$\boxed{\hat{\Sigma}_g = \frac{1}{N-1}\tilde{\mathbf{G}}^\top \tilde{\mathbf{G}}.}$$

Entry (i, j) is the dot product of centered columns i and j, divided by $N - 1$. This evaluates the earlier covariance formula for every pair at once.

Covariance Matrix Properties

The centered-data formula guarantees three properties of our estimate:

- Elements: diagonal entries are nonnegative growth-rate variances; off-diagonal entries are pairwise covariances.
- Symmetry: swapping i and j leaves each product unchanged, so $\hat{\Sigma}_{g,ij} = \hat{\Sigma}_{g,ji}$.
- Positive semidefinite: for any vector $\mathbf{v} \in \mathbb{R}^M$, the centered-data formula gives:

$$\begin{aligned}\mathbf{v}^\top \hat{\Sigma}_g \mathbf{v} &= \frac{1}{N-1} \mathbf{v}^\top \tilde{\mathbf{G}}^\top \tilde{\mathbf{G}} \mathbf{v} \\ &= \frac{\|\tilde{\mathbf{G}}\mathbf{v}\|_2^2}{N-1} \geq 0.\end{aligned}$$

  This is a squared length divided by $N-1 > 0$, so every weighted sum of growth rates has nonnegative sample variance.

Scaling by $\Delta t > 0$ preserves all three properties.

Estimating the Covariance Rate

Under the model, the vector of one-step growth rates $\mathbf{g}$ has covariance $\mathrm{Cov}(\mathbf{g}) = \mathbf{C}/\Delta t$, so growth rates and log returns give equivalent estimates of $\mathbf{C}$:

- From growth-rate data: the covariance rate and L4b volatility estimates are:

$$\hat{\mathbf{C}} = \Delta t\, \hat{\Sigma}_g \qquad \text{(year}^{-1}\text{)},$$

$$\hat{\sigma}_i = \sqrt{\hat{C}_{ii}} = \sqrt{\Delta t}\sqrt{\hat{\Sigma}_{g,ii}} \qquad \text{(year}^{-1/2}\text{)}.$$

- Comparison with log returns: $\mathbf{r} = \Delta t\, \mathbf{g}$ gives:

$$\hat{\Sigma}_r = (\Delta t)^2\, \hat{\Sigma}_g = \Delta t\, \hat{\mathbf{C}} \qquad \text{(dimensionless)}.$$

  For daily data with $\Delta t = 1/252$ years, both inputs give:

$$\hat{\mathbf{C}} = \frac{\hat{\Sigma}_g}{252} = 252\, \hat{\Sigma}_r.$$

These conversions leave correlation unchanged: the common factor cancels.

Example: Covariance and Correlation from Data

Example: Covariance and Correlation from Data

How do firms' growth rates move together?

- Compute the empirical growth-rate covariance matrix and convert it to the GBM covariance rate.
- Check the covariance calculation and compare the implied volatilities with our L4b estimates.
- Interpret the covariance and correlation of a pair of firms.

The estimated covariance tells us how asset growth rates vary together. Next, we decide how much of our investment to assign to each asset.

Portfolio Weights

Once we can simulate several assets together, we need to specify how much to invest in each.

Let $W_0 > 0$ be initial wealth in dollars, and let $\mathbf{w} = (w_1, \dots, w_M)^\top$ contain the fractions invested at time zero.

For a fully invested, long-only portfolio, the weights satisfy:

$$\underbrace{w_i \geq 0}_{\text{long only}}, \qquad \underbrace{\sum_{i=1}^{M} w_i = 1}_{\text{fully invested}}.$$

These constraints define the $(M-1)$-dimensional simplex Δ^{M-1}: a line segment for two assets and a triangle for three.

Buy-and-Hold Wealth

We allow fractional shares and ignore dividends and trading costs. At time zero, we buy and then hold a fixed number of shares of each asset:

$$n_i = \frac{w_i W_0}{S_i(0)}.$$

Portfolio wealth at time t is:

$$\boxed{W_t = \sum_{i=1}^{M} n_i S_i(t) = W_0 \sum_{i=1}^{M} w_i \, \frac{S_i(t)}{S_i(0)}.}$$

The fraction of wealth in asset i then becomes:

$$w_i(t) = \frac{n_i S_i(t)}{W_t}.$$

Share counts remain fixed, but weights change as relative prices move. Maintaining constant weights requires trading to rebalance the portfolio.

Dirichlet Sampling of the Simplex

Draw $\mathbf{w} \sim \mathrm{Dirichlet}(\alpha)$, with concentration parameters $\alpha = (\alpha_1, \ldots, \alpha_M)$, where $\alpha_i > 0$ and $\alpha_0 = \sum_{i=1}^{M} \alpha_i$.

Each draw gives weights that sum to one and are positive with probability one, so the full budget is invested. We use a draw as the initial allocation $\mathbf{w}$. The weight moments are:

$$\mathbb{E}[w_i] = \frac{\alpha_i}{\alpha_0}, \qquad \mathrm{Var}(w_i) = \frac{\alpha_i(\alpha_0 - \alpha_i)}{\alpha_0^2(\alpha_0 + 1)},$$

$$\mathrm{Cov}(w_i, w_j) = -\frac{\alpha_i \alpha_j}{\alpha_0^2(\alpha_0 + 1)}, \qquad i \neq j.$$

These moments describe the sampled portfolio weights. To evaluate portfolio risk, we use the covariance matrix of the firms' growth rates.

How Concentration Shapes the Weights

For the three-firm example, changing α changes how we divide the investment:

- Explore different investment mixes, $\alpha = (1, 1, 1)$: no firm is favored; one portfolio might allocate 70%, 20%, and 10% to the three firms.
- Concentrate the investment, $\alpha = (0.5, 0.5, 0.5)$: favor more uneven allocations, with most money in one or two firms.
- Stay closer to an equal split, $\alpha = (5, 5, 5)$: draws cluster closer to one third of the investment in each firm.
- Favor the first firm, $\alpha = (5, 1, 1)$: it receives about 71% of the budget on average across draws.

Both $(1, 1, 1)$ and $(5, 5, 5)$ average one third per firm across many draws. The larger parameters make individual portfolios closer to that equal split.

Linear Portfolio Growth-Rate Proxy

How do we compare the portfolios that different allocations produce? We summarize each allocation by the mean and variance of its growth rate.

Hold an allocation $\mathbf{w}$ for one period of length Δt, and collect the asset growth rates over that period in $\mathbf{g} = [g_1, \dots, g_M]^\top$. Keeping only terms first order in the log returns $g_i \Delta t$:

$$\frac{1}{\Delta t} \ln\left(\frac{W_{\Delta t}}{W_0}\right) = \frac{1}{\Delta t} \ln\left(\sum_{i=1}^{M} w_i e^{g_i \Delta t}\right) \approx \mathbf{w}^\top \mathbf{g} = g_p.$$

We call g_p the linear growth-rate proxy.

Where is this coming from? use $e^{g_i \Delta t} \approx 1 + g_i \Delta t$, then $\sum_i w_i = 1$, and then $\ln(1 + x) \approx x$.

Estimating Portfolio Growth and Risk

Over the historical observations, the proxy's sample mean and variance are:

$$g'_p = \mathbf{w}^\top \mathbf{g}', \qquad \text{(mean growth rate)}$$

$$\sigma^2_{g,p} = \mathbf{w}^\top \hat{\Sigma}_g \mathbf{w} \geq 0. \quad \text{(variance)}$$

Here, $\mathbf{g}'$ (year^{-1}) and $\hat{\Sigma}_g$ (year^{-2}) come from the covariance estimate. The variance includes each firm's variability and the covariances between firms.

The proxy covers one period. Over time, we use the wealth formula.

From Sampling to Portfolio Choice

A Dirichlet draw gives a candidate allocation. We evaluate its estimated growth and risk before choosing how to invest:

- Sampling: compare many candidate allocations. The lowest-variance draw need not minimize variance over all feasible weights.
- Optimization in L6a: solve for weights that minimize estimated variance, with or without a required mean growth rate.

Example: Portfolio Weights and the Dirichlet Distribution
Explore investment mixes, compare their estimated growth and risk, and follow selected buy-and-hold portfolios using the wealth formula.

Optional Advanced Material

Advanced: Derivation of the Multiple Asset GBM Solution
Why does each asset's drift correction contain only its own variance rate? We extend Itô's lemma to several noise sources and derive the scaling rule.

Advanced: Sampling Error and Covariance Shrinkage
How does estimation error affect portfolio selection? We measure covariance error, combine the sample estimate with a simpler matrix, and compare the resulting minimum-variance portfolios on later data.

Advanced: Rolling Correlations
Are the relationships between firms stable over time? We compare rolling and exponentially weighted correlations and examine changes during periods of high market volatility.

Optional; none of these notebooks is a prerequisite for L6a.

Summary

We modeled correlated asset prices and examined how the fraction invested in each firm affected portfolio growth and risk.

Key takeaways

- Correlated asset prices: We reviewed EMA parameter updates, then combined independent Wiener increments with a loading matrix to model firms whose prices move together.
- Covariance from data: We estimated growth-rate covariance, scaled it by the observation interval to obtain the covariance rate, and used correlation to compare co-movement across firms.
- Portfolio weights: We used Dirichlet draws to explore investment mixes, compared them with a linear growth-rate proxy, and calculated buy-and-hold wealth from fixed share counts.

Next time: we will use minimum-variance optimization to find optimal portfolio weights—the weights that minimize estimated variance under the chosen constraints.